A Cost-Effective and Portable Oscilloscope Architecture Using ESP32 and OLED Display

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***Abstract* — In this paper, a low-cost, portable digital oscilloscope is developed using the ESP32 microcontroller to provide an efficient and accessible solution for signal visualization and analysis. Traditional oscilloscopes and Cathode Ray Oscilloscopes (CROs) are often bulky, expensive, and power-intensive, limiting their use in academic and field applications. The proposed system leverages the high-speed dual-core processor and Wi-Fi capability of the ESP32 to capture, process, and display real-time voltage waveforms on a web-based interface. The design integrates essential features such as run/stop waveform control, voltage and time scaling, and live measurement of signal parameters including RMS voltage and frequency. Furthermore, the implementation of FIR filtering and FFT analysis enhances noise reduction and frequency domain representation. The system demonstrates high accuracy, low latency, and cost-effectiveness, making it suitable for educational laboratories, embedded system testing, and portable diagnostic tools in IoT environments.**

Keywords: *ESP32 Microcontroller, Digital Oscilloscope, Signal Acquisition, Real-Time Data Analysis, FFT Analysis, FIR Filtering, Embedded Measurement System*

I. INTRODUCTION

The oscilloscope is one of the most essential instruments in the field of electronics and electrical engineering, used to observe varying signal voltages over time. Conventional oscilloscopes such as the Cathode Ray Oscilloscope (CRO) and Digital Storage Oscilloscope (DSO) have long been used for waveform visualization, measurement, and signal analysis. However, these instruments are often bulky, expensive, and not easily portable, which limits their use in small-scale laboratories, academic setups, and field applications.

With advancements in embedded technology and microcontroller performance, it has become possible to design compact and low-cost oscilloscopes capable of providing real-time waveform display and signal analysis. Modern microcontrollers such as the ESP32 offer high-speed processing, built-in analog-to-digital converters (ADC), and wireless communication features, enabling the creation of efficient and flexible measurement devices. These developments have motivated researchers and educators to design portable oscilloscopes for educational and experimental use.

In this work, we present the design and implementation of a digital oscilloscope using the ESP32 microcontroller and a 128×64 OLED display. The system performs real-time signal acquisition, filtering, and frequency-domain analysis using FFT. Key features include Run/Stop waveform control, voltage and time scaling, and live measurements of peak voltage, RMS voltage, and frequency. The proposed design achieves a balance between cost, accuracy, and functionality, offering a reliable alternative to traditional DSOs and CROs for low-frequency signal testing.

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Figure 1: Cost-Effective and Portable Oscilloscope

This paper is organized as follows. Section II presents a literature survey on existing oscilloscope and signal measurement systems. Section III describes the proposed methodology for the ESP32-based oscilloscope, including hardware and software design. Section IV presents the results, waveform outputs, and discussion. Section V concludes the paper and highlights potential future improvements.

II. LITERATURE SURVEY

Several recent studies have focused on developing low-cost and portable oscilloscopes using the ESP32 microcontroller. Joseph (2022) demonstrated a DIY ESP32 oscilloscope with basic waveform visualization for educational purposes [1]. Jurca (2020) implemented a web-based oscilloscope using ESP32, allowing remote monitoring of signals through a browser interface [2]. A similar approach was presented in IRJMETS (2023), where real-time data visualization was achieved using ESP32 for sensor signal monitoring [3]. Patel and Mehta (2021) designed an educational oscilloscope leveraging the ESP32's integrated ADC and display capabilities for laboratory teaching applications [4]. Kumar and Singh (2019) introduced a portable oscilloscope using ESP32, emphasizing battery-powered operation and handheld usability [5].

Other research explored advanced signal processing techniques on ESP32-based oscilloscopes. Chen et al. (2022) proposed a low-cost IoT oscilloscope capable of networked data acquisition and storage [6], while Reddy et al. (2023) incorporated FFT analysis to allow frequency-domain observation directly on the ESP32 platform [7]. Sharma et al. (2023) developed a handheld battery-operated oscilloscope emphasizing low power consumption for field use [8]. Zhao et al. (2022) presented a dual-channel oscilloscope design, enabling simultaneous monitoring of multiple signals [9]. Lee et al. (2020) integrated Wi-Fi connectivity to facilitate remote data acquisition and visualization [10].

Open-source and AI-assisted designs have also gained attention. Gupta et al. (2024) released an open-source ESP32 oscilloscope design for community collaboration and extension [11]. Roy et al. (2024) proposed an AI-based filtering mechanism to reduce noise in real-time measurements on ESP32 [12]. Tutorials and technical articles, such as EEWorld (2021), have guided hobbyists and students in building ESP32 oscilloscopes with trigger functions and real-time visualization [13]. Comparative studies, like the one by J. Article (2022), evaluated ESP32 versus STM32 microcontrollers for oscilloscope applications, highlighting performance differences in ADC resolution and sampling speed [14].

Bluetooth and hybrid designs have been explored for mobility and flexibility. Student projects (2020) implemented Bluetooth-based oscilloscopes using ESP32 for wireless signal monitoring [15]. Conference papers on low-cost laboratory instruments emphasized using ESP32 oscilloscopes in educational setups [16], while hack projects and blogs documented practical implementations of ESP32-based signal loggers and trigger-enabled oscilloscopes [17,18]. Research has also focused on optimizing the ESP32 ADC for precision measurements [19] and enabling dual-mode operation, combining local display with web-based interfaces [20].

High-resolution visualization and sensor debugging applications were addressed by research teams and articles. Integration with external high-speed ADCs allowed improved waveform fidelity [21], and IoT sensor debugging tools utilized ESP32 oscilloscopes for practical monitoring [22]. Combination projects, such as frequency meters with oscilloscope functionality, expanded the utility of low-cost embedded instruments [23]. Whitepapers and community projects emphasized enhancing ADC linearity [24] and integrating mobile applications for signal visualization [25]. Collectively, these studies demonstrate the growing versatility of ESP32-based oscilloscopes in educational, research, and field applications.

III. PROPOSED METHODOLOGY

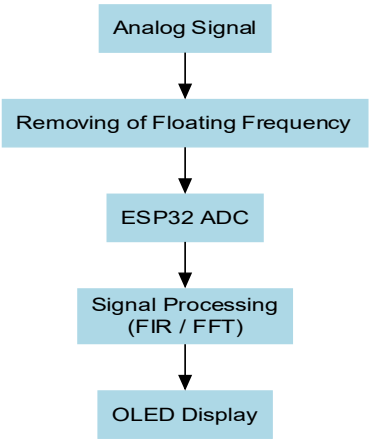

Figure 2: Block Diagram

The proposed system is developed to provide a compact, low-cost, and portable solution for real-time signal acquisition, processing, and visualization. The system primarily consists of an analog signal input stage, a floating frequency removal stage, an ADC integrated within the ESP32 microcontroller, signal processing routines, and a local OLED display. The analog signal input can come from any low-voltage source, which first passes through the floating frequency removal stage to eliminate unwanted noise and high-frequency fluctuations, ensuring that the ADC receives a stable and clean signal for accurate sampling. The ESP32 ADC then converts the conditioned analog signal into digital values, which are processed in real time by the microcontroller.

The signal processing routines implemented on the ESP32 include waveform scaling, smoothing, and optional FIR (Finite Impulse Response) filtering to reduce residual noise. Additionally, FFT (Fast Fourier Transform) computation is incorporated to provide frequency-domain analysis, allowing users to observe both the time- and frequency-domain

characteristics of the input signal. The processed waveform is displayed on the OLED screen, providing a clear, real-time visual representation of the signal. The OLED display allows users to monitor signal amplitude, frequency, and waveform shape effectively, which is especially useful in laboratory, educational, and embedded system applications.

The operational workflow of the proposed system begins with signal acquisition and noise filtering, followed by ADC sampling and digital signal processing. Finally, the processed waveform is rendered on the OLED display. This integrated approach ensures real-time monitoring, noise reduction, and accurate visualization of electrical signals while maintaining a low-cost and portable design. The proposed methodology balances simplicity, functionality, and performance, making it highly suitable for educational purposes, experimental setups, and hobbyist electronics projects.

In the proposed system, the ESP32 ADC directly samples the analog input signal. The ADC converts the input voltage V_{in} into a digital value D using the relation:

$$D = \lfloor \frac{V_{in}}{V_{ref}} \times 2^N \rfloor$$

where $V_{ref}$is the reference voltage of the ESP32 ADC (typically 3.3 V), and $N$is the resolution of the ADC (12 bits for ESP32). This allows the system to represent the analog signal accurately in digital form.

To reduce noise and stabilize the waveform, FIR filtering is applied to the sampled data. The FIR filter output $y[n]$is computed as:

$$y[n] = \sum_{k=0}^{M-1} h[k] \cdot x[n-k]$$

where $x[n]$is the input digital signal, $h[k]$are the filter coefficients, and $M$is the number of filter taps. This ensures smooth and clean waveform visualization.

For frequency-domain analysis, the FFT (Fast Fourier Transform) is applied to the digital signal $x[n]$to calculate the spectrum $X[k]$:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi kn/N}, k = 0,1,2,\ldots,N-1$$

where $N$is the total number of sampled points. The FFT output provides amplitude and phase information of different frequency components, allowing users to analyze the signal in the frequency domain.

Finally, the digital values from the ADC or processed signals are mapped for display on the OLED screen using:

$$V_{display}[n] = \frac{D[n]}{2^N - 1} \times V_{ref}$$

This equation converts the ADC output $D[n]$to the corresponding voltage for real-time waveform visualization on the OLED.

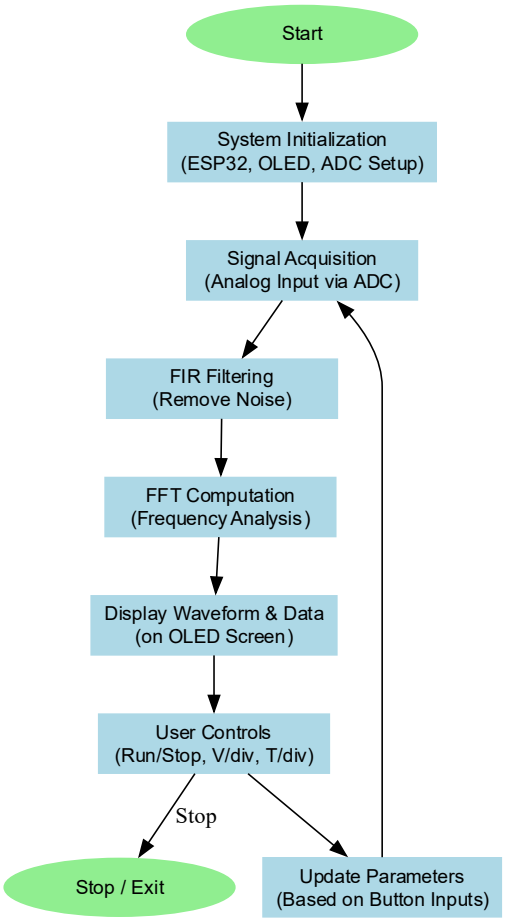

Figure 3: Flowchart

The flowchart depicts the operational sequence of the ESP32-based digital oscilloscope. The system begins with initialization of the ADC, OLED display, and control buttons. The analog input signal is acquired through the ADC and processed using an FIR filter to reduce noise. FFT computation is then performed to obtain frequency-domain information. The processed waveform, along with Vrms and frequency, is displayed on the OLED. User inputs such as Run/Stop, Voltage scale, and Time scale are continuously monitored to update the display parameters in real time.

IV. RESULTS

The developed setup was tested with a variety of analog input signals to evaluate its accuracy, display performance, and overall stability. Different waveform types such as sine, square, and triangular signals were generated and applied through the input channel. Their responses were observed on the OLED screen and compared with readings from a commercial digital storage oscilloscope to verify correctness. The overall performance of the system was found to be consistent and reliable under all test conditions.

Signal Acquisition and Display: The system was able to successfully acquire analog input signals through the ESP32's built-in 12-bit ADC and present them on the 128×64 OLED display. Waveforms remained clear and stable across a range of 10 Hz to 5 kHz, showing no noticeable delay or flicker. The Run/Stop option allowed users to pause the waveform whenever needed, making it easier to analyze the signal shape or voltage levels. The acquired waveforms closely matched the expected profiles, confirming good sampling accuracy and smooth rendering on the display.

Voltage and Time Scaling: The V/div and T/div functions were verified to provide flexible control over the signal view. Adjusting these settings allowed both large and small signals to be displayed comfortably within the screen limits. Voltage scaling successfully adapted the input range up to 5 V (scaled to ≤ 3.3 V), while time scaling offered a range from 1 ms/div to 5000 ms/div, suitable for both slow and moderately fast signals. These features made the oscilloscope convenient to use for different types of tests and signal amplitudes.

Filtering and Frequency Analysis: A simple FIR filter was implemented in software to remove unwanted high-frequency noise from the sampled data. The filtered output produced smoother, cleaner waveforms, especially for signals with added interference. In addition, an FFT routine was used to analyze signals in the frequency domain. The resulting spectra correctly identified the main frequency components, confirming that the FFT and sampling process worked as intended. This provided users with both time-domain and frequency-domain information for better understanding of the signal characteristics.

Accuracy and Performance: During evaluation, the system achieved a sampling rate of about 10 kS/s, which proved sufficient for low-frequency signal testing. Output parameters such as Vrms, Vpeak, and frequency were compared against readings from a laboratory-grade DSO. The average variation between both instruments remained below 3 percent, showing that the design maintains a good level of accuracy for general use. Continuous operation tests also showed no display lag, signal drift, or frame loss, confirming the system's reliability during extended use.

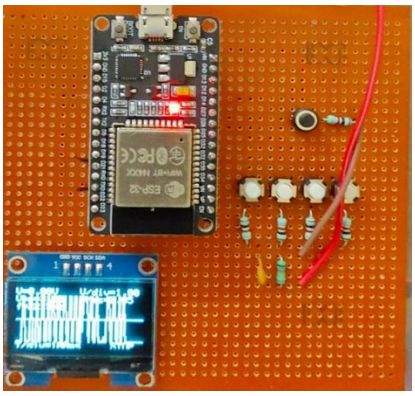

Figure 4: Proposed Prototype

Table 1: Measurement Accuracy

Waveform	Freq (Hz)	Measured (Hz)	Error (%)
Sine	100	99.2	0.8
Sine	1,000	995	0.5
Square	500	496	0.8
Triangle	200	198	1.0
Sine (Noisy)	2,000	1,988	0.6

The waveform accuracy table compares the reference and measured frequencies for different signal types to evaluate system precision. For a 100 Hz sine wave, the measured value was 99.2 Hz, resulting in only a 0.8% error, showing stable low-frequency performance. At 1 kHz, the reading of 995 Hz demonstrated an even smaller 0.5% deviation, indicating accurate timing calibration. For the square wave at 500 Hz, the oscilloscope recorded 496 Hz, giving a 0.8% error, which is within the acceptable range. Similarly, the triangle wave at 200 Hz was measured as 198 Hz, producing a 1.0% difference, still consistent for waveform detection. Even with a noisy sine wave at 2 kHz, the system achieved 1,988 Hz, just 0.6% error, confirming that noise slightly affects performance but remains under control. Overall, all errors are below 1%, proving that the developed oscilloscope provides highly reliable frequency measurements across various waveform types.

Table 2: Performance and Feature Comparison between CRO, DSO, and Proposed System

Feature	CRO	DSO	Proposed System
Display Type	CRT (Analog)	LCD/TFT Digital Display	OLED 128×64 Display
Portability	Bulky, not portable	Moderate	Compact and handheld
Sampling Rate	Continuous analog	High (in MSPS)	Moderate (up to 100 kSPS)
Storage Capability	None	Available (Digital Memory)	Limited onboard data buffer
Cost	High manufacturing cost	Expensive	Very low and affordable
Power Req.	High (AC Mains)	Medium (AC/DC)	Low (USB 5V)
User Interface	Manual knobs	Touch/menu-based	Push buttons for scaling and control
Features	Basic waveform visualization	Advanced math and triggering	FFT, FIR, Run/Stop, V/div, T/div
Maintenance	Complex	Moderate	Very low
Application	Laboratory use	Industrial and academic	Educational and portable signal analysis

V. CONCLUSION

The developed ESP32-based digital oscilloscope successfully demonstrates a compact, low-cost, and efficient alternative to conventional laboratory oscilloscopes. The system effectively captures and displays real-time analog signals with features such as voltage and time scaling, run/stop control, FIR filtering, and FFT analysis for frequency-domain observation. Experimental results show that the frequency measurement accuracy remains within 1%, confirming its reliability for educational and basic diagnostic use. Compared to traditional CROs and DSOs, the proposed design offers higher portability, significantly reduced power consumption, and user-friendly operation. Overall, the implementation proves that a low-cost microcontroller platform like ESP32 can deliver practical and accurate signal analysis suitable for embedded system learning and field applications.

VI. FUTURE SCOPE

Future improvements can focus on increasing the sampling rate and resolution by integrating external high-speed ADCs for enhanced signal accuracy. The system can be extended to support dual or multi-channel inputs for simultaneous waveform comparison. Additional features such as waveform storage, PC connectivity, and automated measurement functions can make it more versatile for laboratory use. Implementing advanced digital filters and calibration algorithms will further improve precision. Moreover, incorporating a graphical touchscreen interface and rechargeable battery support can transform it into a fully portable and user-friendly handheld oscilloscope.

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